

Curriculum Vitae

Hao Wang, Ph.D.
Associate Professor
Shanghai Eastern Youth Scholar
First Prize of the Outstanding Teaching Award of Shanghai, 2022
School of Information Science and Technology
ShanghaiTech University

Biographical Information

Business Address

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Educational History

Ph.D. Lehigh University, September 2009–May 2015
Industrial and Systems Engineering
Dissertation title: *Matrix-free Methods for Large Scale Nonlinear Optimization and Rapid Infeasibility Detection*
.....advisor: Prof. Frank E. Curtis
.....committee: Prof. Tamas Terlaky
Prof. James V. Burke
Prof. Katya Scheinberg
M.S. Beihang University, September 2007–July 2009
Mathematics and Applied Mathematics
B.S. Beihang University, September 2003–July 2007
Mathematics and Applied Mathematics

Employment History

Associate Professor	ShanghaiTech University, 2023–present School of Information Science and Technology
Assistant Professor	ShanghaiTech University, 2016–2023 School of Information Science and Technology
Data Scientist	GroupM Corporation, January 2015–February 2016 Research and Development
Research Assistant Intern	Mitsubishi Corporate, September 2014–November 2014 Mitsubishi Electric Research Laboratory (MERL, MA), Data Analytic Group
Research Assistant Intern	ExxonMobil Corporate, June 2012–August 2012 Corporate Strategic Research Laboratory (NJ)
Intern	Honeywell Corporate, May 2009–July 2009 Technology Solutions Laboratory (Beijing)

Working Papers

Papers under review

1. Yinhao Zhao, Haoyu He, Chuanqi Ma, Hao Wang, “A Unified Fractional Regularization Framework for Sparse Recovery,” submitted to *Journal of Scientific Computing*, under first review.
2. Aike Yang, Hao Wang, “Stochastic Trust-Region Methods for Over-parameterized Models.” submitted to *Optimization Methods and Software*.
3. Haoyu He, Jinyu Zhuang, Haoran Chu, Shuhang Yu, J, T AI Group, Hao Wang, Kunpeng Han, “From Synthetic to Native: Benchmarking Multilingual Intent Classification in Logistics Customer Service”, [arXiv:2603.23172](#)
4. Haoyu He, Hao Wang*, Jiashan Wang, Hao Zeng, “Effective Sparsity: A Unified Framework via Normalized Entropy and the Effective Number of Nonzeros,” submitted to *Journal of Machine Learning Research*, [arXiv:2603.13826](#).
5. Hao Wang, Xiangyu Yang, Yichen Zhu, “Alternating Subspace Newton and Iteratively Reweighted ℓ_1 Algorithms for Nonconvex Sparse Optimization,” submitted to *Informa Journal on Optimization*, under second review.
6. Kexin Li, Luwei Bai, Xiao Wang, Hao Wang*, Anderson Acceleration in Nonsmooth Problems Local Convergence via Active Manifold Identification, submitted to *IMA Journal of Numerical Analysis*.
7. Hongying Liu, Hao Wang*, Haoran Chu, Yibo Wu, Towards Convexity in Anomaly Detection: a New Formulation of SSLM with Unique Optimal Solutions, submitted to *Journal of Machine Learning Research*, under second review.

Journal Articles

1. Xiangyu Yang, Hao Wang*, Yichen Zhu, Xiao Wang, “Minimization Over the Nonconvex Sparsity Constraint Using A Hybrid First-order method,” accepted by **Journal of Scientific Computing**.
2. Yaohua Hu, Hao Wang, Xiaoqi Yang, “ ℓ_{1-2} Regularization for Sparse Optimization: Consistency and Global Convergence,” accepted by **Mathematical Operations Research**.
3. Luwei Bai, Yaohua Hu, Hao Wang*, Xiaoqi Yang, “Avoiding Strict Saddle Points of Nonconvex Regularized Problems,” accepted by **SIAM Journal on Optimization**.
4. Qiankun Shi, Xiao Wang*, Hao Wang, “A momentum-based linearized augmented Lagrangian method for nonconvex constrained stochastic optimization,” published online on **Mathematical Operations Research**, <https://doi.org/10.1287/moor.2022.0193>.
5. Wengqing Ouyang, Yuncheng Liu, Ting Kei Pong, Hao Wang, “Kurdyka-Lojasiewicz Exponent via Hadamard Parameterization,” **Journal SIAM Journal on Optimization**, vol. 35, no. 1, pp. 62–91, August, 2025.
6. Hao Wang, Ye Wang, Xiangyu Yang, “Efficient Active Manifold Identification via Accelerated Iteratively Reweighted Nuclear Norm Minimization,” **Journal of Machine Learning Research**, vol. 25, no. 319, pp. 1–44, 2024.
7. Tiange Li, Xiangyu Yang*, Hao Wang, “A Fast Solver for Structured Optimization with Nonconvex $\ell_{q,p}$ Regularization,” accepted by **Pacific Journal of Optimization**.
8. Hao Wang*, Yining Gao, Jiashan Wang, and Hongying Liu, “Constrained Optimization Involving Nonconvex ℓ_p Norms: Optimality Conditions, Algorithm and Convergence”, **Pacific Journal of Optimization**.

9. Xiangyu Yang, Hao Wang*, Jiashan Wang, “Towards an efficient approach for the nonconvex lp ball projection: algorithm and analysis”, **Journal of Machine Learning Research**, vol. 23, no. 101, pp. 1–31, April, 2022.
10. Hao Wang*, Hao Zeng, Jiashan Wang, “An Extrapolated Iteratively Reweighted ℓ_1 Method with Complexity Analysis”, **Computational Optimization and Applications** no. 83, pp. 967–997, September, 2022.
11. Hao Wang*, Hao Zeng, Jiashan Wang, “Convergence Rate Analysis of Proximal Iteratively Reweighted ℓ_1 Methods for ℓ_p Regularization Problems”, **Optimization Letters**, vol. 17, no. 2, pp. 413–435, 2023.
12. Yuyan Zhou, Yang Liu*, Ming Li, Qingqing Wu, Hao Wang. “Joint Sensor Selection, Beamforming and Phase Control in Reconfigurable Intelligent Surface Aided IoT Networks”, **IEEE Wireless Communications Letters**, vol. 11, no. 2, pp. 401–405, February, 2022.
13. Hao Wang*, Fan Zhang, Yaohua Hu, Yuanming Shi. “Nonconvex and Nonsmooth Sparse Optimization via Adaptively Iterative Reweighted Methods”, **Journal of Global Optimization**, vol. 81, no. 3, pp. 717–748, October, 2021.
14. Hejie Wei, Zhihai Qu, Xuyang Wu, Hao Wang, Jie Lu, “Decentralized Approximate Newton Methods for Convex Optimization on Networked Systems,” vol. 8, no. 3, pp. 1489–1500, September, 2021, **IEEE Transactions on Control of Network Systems**.
15. Xiangyu Yang, Hao Wang*, Yuanming Shi, Jun Lin, “A Proximal Iteratively Reweighted Approach for Efficient Network Sparsification,” vol. 71, no. 1, pp. 185–196, April, 2022, **IEEE Transactions on Computers**.
16. Hao Wang*, Hao Zeng, Jiashan Wang, Qiong Wu, “Relating ℓ_p regularization and reweighted ℓ_1 regularization”, **Optimization Letters**, vol. 15, no. 8, pp. 2639–2660, January, 2021.
17. Hongying Liu, Hao Wang*, Mengmeng Song. “Projections onto the Intersection of a One-Norm Ball or Sphere and a Two-Norm Ball or Sphere,” **Journal of Optimization Theory and Applications**, vol. 187, pp. 520–534, October, 2020.
18. James V. Burke, Frank E. Curtis, Hao Wang*, Jiashan Wang. “Inexact Sequential Quadratic Optimization With Penalty Parameter Updates Within The QP Solver”, **SIAM Journal on Optimization**, vol. 30, no. 3, 2020.
19. Xiangyu Yang, Sheng Hua, Yuanming Shi, Hao Wang, Jun Zhang, B. Letaief, Khaled, “Sparse Optimization for Green Edge AI Inference,” **Journal of Communications and Information Networks**, vol. 5, no. 1, 2020.
20. Zhihua Zhao, Hao Wang*, Xiangyu Yang, Fengmin Xu, “CVaR-Cardinality Enhanced Indexation Optimization with Tunable Short-selling Constraints”, **Applied Economics Letters**, March, 2020.
21. Hao Wang*, Fan Zhang, Jiashan Wang, Yuyang Rong. “An Inexact First-Order Method for Constrained Nonlinear Optimization”, **Optimization Methods and Software**, pp. 1–34, January, 2020.
22. Fan Zhang, Hao Wang*, Jiashan Wang, Kai Yan, “Inexact Primal-Dual Gradient Projection Methods for Nonlinear Optimization on Convex Set”, **Optimization**, pp. 1–27, December, 2019.
23. Yang Liu, Jing Li, Hao Wang. “Robust Linear Beamforming in Wireless Sensor Networks”, **IEEE Transactions on Communications**, vol. 67, no. 6, pp. 4450–4463, June, 2019.
24. Qiong Wu, Fan Zhang, Hao Wang*, Jun Lin, Yang Liu. “Parameter-free ℓ_p -Box Decoding of LDPC Codes”, **IEEE Communications Letters**, vol. 22, no. 7, pp. 1318–1321, July, 2018.

25. James V. Burke, Frank E. Curtis, Hao Wang*, J. Wang. “Iterative Reweighted Linear Least Squares for Exact Penalty Subproblems on Product Sets”, **SIAM Journal on Optimization**, vol. 25, no. 1, pp. 261–294, 2015.
26. James V. Burke, Frank E. Curtis, Hao Wang*. “A Sequential Quadratic Optimization Algorithm with Rapid Infeasibility Detection”, **SIAM Journal on Optimization**, vol. 24, no. 2, pp. 839–872, 2014.
27. Hao Wang*, Hongying Liu, Yong Xia. “Two-point Step Size Iterative Soft-thresholding Method for Sparse Reconstruction”, **International Journal of Computer Mathematics**, vol. 88, no. 12, pp. 2527–2537, April, 2010.
28. Hao Wang*, Hongying Liu, Yong Xia. “Two-step Version of the Fixed Point Continuation Method for Sparse Reconstruction”, **Frontiers of Mathematics in China**, vol. 5, no. 3, pp. 575–588, May, 2010.

Conference Articles

1. Yanjie Chen, Likun Cai, Wei Cheng, Hao Wang, “Super-Resolution Coding Defense against Adversarial Examples”, ICMR 2020: Proceedings of the 2020 International Conference on Multimedia Retrieval. June 2020 pp.189–197.
2. S. Hua, X. Yang, K. Yang, G. Yin, Y. Shi and H. Wang, “Deep Learning Tasks Processing in Fog-RAN,” 2019 IEEE 90th Vehicular Technology Conference (VTC2019-Fall), Honolulu, HI, USA, 2019, pp. 1–5.
3. T. Jiang, X. Yang, Y. Shi and H. Wang, “Layer-wise Deep Neural Network Pruning via Iteratively Reweighted Optimization,” ICASSP 2019–2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Brighton, United Kingdom, 2019, pp. 5606–5610.
4. Jie Lu, Xuyang Wu, Hejie Wei, Hao Wang, “Distributed Second-Order Methods for Convex Optimization Over Networks,” the proceedings of the 15th International Conference on Control, Automation, Robotics and Vision, (ICARCV), Singapore, 2018.
5. Fan Zhang, Qiong Wu, Hao Wang, Yuanming Shi. “Topological interference alignment via generalized low-rank optimization with sequential convex approximations,” in Proc. IEEE Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC), Kalamata, Greece, Jun. 2018.
6. Fan Zhang, Qiong Wu, Hao Wang, Yuanming Shi. “ ℓ_p -Box Optimization for Green Cloud-RAN via Network Adaptation”, in Pro. IEEE Veh. Technol. Conf. (VTC), Chicago, USA, Aug. 2018.
7. Q. Wu, F. Zhang, H. Wang and Y. Shi, “Generalized Low-Rank Matrix Completion via Nonconvex Schatten p -Norm Minimization,” 2018 IEEE 88th Vehicular Technology Conference (VTC-Fall), Chicago, IL, USA, 2018, pp. 1-5.

Research Funding

1. “Optimization Methods for Nonconvex Low-Rank Matrix Regularization with Applications in Large-Scale Computing,” General Program of the National Natural Science Foundation of China (NSFC), Project No. 12571326, Funding: 430,000 CNY, Duration: 01/2026–12/2029, Role: **Principal Investigator (PI)**.
2. “High Performance Algorithms for Large-Scale Nonconvex Regularized Machine Learning,” Natural Science Foundation of Shanghai, Project No. 21ZR1442800, Funding: 200,000 CNY, Duration: 07/2021–06/2024, **Principal Investigator (PI)**.

3. “Cubic Regularization Algorithms for Large-Scale Machine Learning Problems,” Young Scientists Fund of the National Natural Science Foundation of China (NSFC), Project No. 12001367, Funding: 240,000 CNY, Duration: 07/2020–06/2023, **Principal Investigator (PI)**.
4. “Model Reduction Based Tensor Recovery Methods for Partial Differential Equations with High-Dimensional Random Inputs,” National Natural Science Foundation of China (NSFC), Project No. 12071291, Funding: 510,000 CNY, Duration: 01/2021–12/2024, Role: Co-Principal Investigator (Co-PI) with Prof. Qifeng Liao.
5. “Data Driven Physics-Informed Uncertainty Quantification for Atmospheric Aerosols,” Shanghai 2020 Scientific and Technological Innovation Program, Project No. 20JC1414300, Funding: 500,000 CNY, Duration: 10/2020–09/2023, Role: Co-Principal Investigator (Co-PI) with Prof. Qifeng Liao.
6. “Shanghai Eastern Youth Scholar Award,” Funding: 400,000 CNY, Duration: 2017–2020, Sponsor: Shanghai Municipal Education Commission.
7. “Research on Internet Resource Management and Optimization Methods Balancing Fairness and Efficiency,” **General Program** of the National Natural Science Foundation of China (NSFC), Project No. 60970104, Funding: 90,000 CNY, Duration: 01/2010–12/2012, Host: Xu Ke (Tsinghua University), Role: Participant.

Graduate Student Advisees

Ph.D. Student Advisees

- Fan Zhang Fall 2016–Spring 2021
Dissertation: *Inexact First-Order Methods for Constrained Nonlinear Optimization*
Position: Huawei → CATL → Apple
- Xiangyu Yang Fall 2017–Spring 2020
Dissertation: *Nonconvex Regularizer Constrained Optimization in Machine Learning: Algorithm, Analysis and Applications*
Position: Assistant Professor, Math Department, Henan University

Master Student Advisees

- Qiong Wu, Excellent Graduate Student Award Fall 2016–Spring 2019
Thesis: *Research on Norm-based Optimization Problems and Algorithms*
Position: Cainiao
- Yanjie Chen Fall 2017–Spring 2020
Thesis: *Defense Against Adversarial Examples Based on Super-Resolution Coding*
Position: Algorithm Engineer, Audio & Music Intelligence, Bytedance-Speech
- Xin Deng Fall 2018– Spring 2021
Thesis: *A Hybrid First-Order Method for Nonconvex ℓ_p -ball Constrained Optimization*
Position: Cainiao → ByteDance → Ant Group → Hellobike
- Wei Jiang Fall 2018–Spring 2021
Thesis: *Nonconvex ℓ_p -ball Constrained Minimization via Projected Gradient Descent*
Position: Algorithm Engineer in Rongshu Tech.
- Hao Zeng Fall 2019– Spring 2022
Thesis: *Algorithms and Theories for ℓ_p Regularization Problems*
Position: Algorithm Engineer in Operations Research and Optimization, Alibaba

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- Yining Gao Fall 2019–Spring 2022
 Thesis: *Constrained Optimization Involving Nonconvex ℓ_p Norms: Optimality Conditions, Algorithm and Convergence*
 Position: China Banking and Insurance Regulatory Commission (Jiaxing, Zhejiang)
 - Song Mao Fall 2019–Spring 2022
 Thesis: *An efficient primal-dual Newton method for projections onto $\ell_{1,\infty}$ norm ball*
 Position: Shanghai AI Lab
 - Ye Wang Fall 2020–Spring 2023
 Thesis: *Algorithm Design and Theoretical Analysis for Solving Schatten- p Norm Regularization Problem*
 Position: Algorithm Engineer, Smart Logic
 - Qiankun Shi Fall 2020–Spring 2023
 Thesis: *First-order Penalty Methods for Large Scale Constrained Optimization*
 Position: PhD candidate at Sun Yat-sen University.
 - Tiange Li Fall 2020–Spring 2023
 Thesis: *Research on the acceleration strategy of group sparse optimization*
 Position: Algorithm Engineer, Qboson
 - Kexin Li Fall 2021–Fall 2024
 Thesis: Study on the Iteratively Reweighted Algorithms with Anderson Acceleration
 Position: PhD, Industrial Engineering, University of Arkansas
 - Luwei Bai Fall 2021–Fall 2024
 Thesis: Avoiding strict saddle points of nonconvex regularized problems
 Position: PhD, Industrial Engineering, Arizona State University
 - Chuanqi Ma Fall 2021–Spring 2025
 Thesis: Sparse Optimization and Algorithmic Research Based on Fractional Regularization
 Position: HotMaxx, Shanghai
 - Shiji Zuo Fall 2022–Spring 2025
 Thesis: Research on the Adaptive Stochastic Linearized Augmented Lagrangian Method
 Position: PhD, Math, Rensselaer Polytechnic Institute
 - Yichen Zhu Fall 2022–Spring 2025 (expected)
 Thesis: Second Order Method for Sparsity Regularizations
 Position: Postal Savings Bank of China, Shanghai
 - Aike Yang Fall 2023–Spring 2026 (expected)
 - Zirui Han Fall 2023–Spring 2026 (expected)
 - Haoran Chu Fall 2023–Spring 2026 (expected)
 - Ken Chen Fall 2024–Spring 2027 (expected)
 - Haoyu He Fall 2024–Spring 2027 (expected)
 - Yuqi Yang Fall 2024–Spring 2027 (expected)
 - Yinhao Zhao Fall 2024–Spring 2027 (expected)
 - Yiran Liu Fall 2024–Spring 2027 (expected)

Teaching

- First Prize (Team Award), the Outstanding Teaching Achievement Award of Shanghai, Shanghai Municipal Education Commission, 2022

SI152 *Numerical Optimization* (Undergraduate Selective Course), 2020–present.

SI140 *Probability and Statistics* (Undergraduate Core Course), 2016–2020.

CS282 *Machine Learning* (Undergraduate and Graduate Course), 2016–present.

Awards

- First Prize, the Outstanding Teaching Achievement Award of Shanghai, Shanghai Municipal Education Commission, 2022
- Shanghai Eastern Youth Scholar, Shanghai Municipal Education Commission, 2017
- P. C. Rossin Doctoral Fellowship, Rossin College at Lehigh University, 2013
- Honeywell Innovators Scholarship, Honeywell Corporate, 2009

Referee for Professional Journals

Mathematical Programming

SIAM Journal on Optimization

IEEE Signal Processing Letter

Automatica

European Journal on Operational Research

Mathematics of Operations Research

Computer Methods in Applied Mechanics and Engineering

Numerical Algorithms

Optimization Letters

Conference Presentations

- “Anderson Acceleration in Nonsmooth Problems: Local Convergence via Active Manifold Identification,” ICCOPT, Los Angeles, California, July, 2025.
- “Towards Convexity in Anomaly Detection: A New Formulation of SSLM with Unique Optimal Solutions,” EURO, Leeds, UK, June, 2025.
- “Alternating Subspace Newton and Iteratively Reweighted l1 Algorithms for Nonconvex Sparse Optimization,” MOPTA, Ponta Delgada, Azores, Portugal, June, 2025.
- “Avoiding strict saddle points of nonconvex regularized problems”, EURO, Copenhagen, August, 2024.
- “Iteratively Reweighted Nuclear Norm Methods for Low-Rank Minimization with Rank Identification”, EUROPT, Lund, Sweden, July 2024.

- “An Inexact First-order Method for Constrained Nonlinear Optimization”, ICCOPT, Berlin, August 2019.
- “Space-reducing algorithms for solving l_p regularization”, Informs 2019, Seattle, USA, November 2019.
- “Updating the Penalty Parameter”, Beijing Society of Operations Research, Beijing, December 2018.
- “Iteratively Reweighted Algorithms for Nonconvex Regularization”, Chinese Society of Optimization, Overall Planning and Economical Mathematics, Xinyang, August, 2018.
- “A Dynamic Penalty Parameter Updating Strategy for Matrix-free Sequential Quadratic Optimization”, ISMP 2018, Bordeaux, France, July 2018.
- “An Iteratively Reweighted Algorithm for Nonconvex Regularization”, EURO 2018, Valencia, Spain, July 2018.
- “A Dynamic Penalty Parameter Updating Strategy for Matrix-free Sequential Quadratic Optimization”, ICCOPT 2016, Tokyo, Japan, August 2016.
- “An Iterative Reweighting Algorithm for Exact Penalty Subproblems”, SIAM Conference on Optimization, San Diego, California, May 2014.
- “Matrix-free Solvers for Exact Penalty Subproblem”, INFORMS Annual Meeting, Minneapolis, Minnesota, October 2013.
- “Matrix-free Solvers for Systems of Inclusions”, ICCOPT 2013, The Fourth International Conference on Continuous Optimization, Lisbon, Portugal, July 2013.
- “A Penalty-SQP Method with Rapid Infeasibility Detection”, INFORMS Annual Meeting, Phoenix, Arizona, October 2012.
- “A Sequential Quadratic Optimization Algorithm with Rapid Infeasibility Detection”, INFORMS Optimization Society Conference, Coral Gables, Florida, USA, February 2012.
- “A Two-point Step Size Iterative Soft-thresholding Method for Sparse Reconstruction”, COR@L seminar, Lehigh University, February 2011.
- “Gradient-type Methods for Linear Inverse Problems with Sparsity Constraints”, The Second International Conference on Nonlinear Programming with Applications 2008 (NPA2008), Beijing, May 2008.

Seminars at Universities and Research Centers (Invited)

- “Iteratively reweighted ℓ_1 methods for ℓ_p regularization: properties, complexity and acceleration”, University of Chinese Academy of Sciences, October, 2021.
- “A hybrid method for solving the ℓ_p ball constrained problems”, Henan University, October, 2021.
- “A hybrid method for solving the ℓ_p ball constrained problems”, Shanghai Jiaotong University, August, 2021.
- “An efficient ℓ_p ball projection method”, Harbin Institute of Technology, July, 2021.
- “Solving the ℓ_p ball constrained problems”, Beihang University, June, 2021.
- “Inexact optimization algorithms using Fenchel-Rockafellar duality”, Xiamen University, July, 2019.
- “Constrained optimization using Fenchel-Rockafellar duality”, North Minzu University, December 2018.

- “Inexact gradient projections methods”, Xi’an Jiaotong University, December 2018.
- “Linear and nonlinear optimization models in scheduling”, Youhualin Corporate, May, 2018.
- “A penalty-SQP method with rapid infeasibility detection”, Hebei University of Technology, November, 2017.
- “Large-scale optimization with applications to machine learning”, Tongji University, September 2017.